Verifying Inverses & Writing Inverses — Practice

Name: _______________ Period: ______

Part 1: Verify that the functions are inverses by showing $f(g(x)) = x$ AND $g(f(x)) = x$.

1. $f(x) = x^3 + 5$ and $g(x) = \sqrt[3]{x-5}$

$f(g(x)) =$ $g(f(x)) =$

Are they inverses? ______________

2. $f(x) = x^3 - 4$ and $g(x) = \sqrt[3]{x} + 4$

$f(g(x)) =$ $g(f(x)) =$

Are they inverses? ______________

3. $f(x) = \sqrt[3]{x+4} - 1$ and $g(x) = (x+1)^3 - 4$

$f(g(x)) =$ $g(f(x)) =$

Are they inverses? ______________

4. $f(x) = 2x^3$ and $g(x) = \sqrt[3]{2x}$

$f(g(x)) =$ $g(f(x)) =$

Are they inverses? ______________

5. $f(x) = (x-3)^3 + 7$ and $g(x) = \sqrt[3]{x-7} + 3$

$f(g(x)) =$ $g(f(x)) =$

Are they inverses? ______________

6. $f(x) = -2(x+1)^3 + 4$ and $g(x) = \sqrt[3]{\dfrac{4-x}{2}} - 1$

$f(g(x)) =$ $g(f(x)) =$

Are they inverses? ______________

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Name: ________________

Part 2: Write the inverse using transformations. Identify the transformations, reverse them, then write $f^{-1}(x)$.

7. $f(x) = (x+4)^3 - 3$

Parameter	Original Transformation	Inverse Transformation
$h =$ ______		
$k =$ ______		

$f^{-1}(x) =$ ______________________________

8. $f(x) = \sqrt[3]{x-6} + 2$

Parameter	Original Transformation	Inverse Transformation
$h =$ ______		
$k =$ ______		

$f^{-1}(x) =$ ______________________________

9. $f(x) = 3(x-1)^3 + 4$

Parameter	Original Transformation	Inverse Transformation
$a =$ ______		
$h =$ ______		
$k =$ ______		

$f^{-1}(x) =$ ______________________________

10. $f(x) = -\sqrt[3]{x+5} - 2$

Parameter	Original Transformation	Inverse Transformation
$a =$ ______		
$h =$ ______		
$k =$ ______		

$f^{-1}(x) =$ ______________________________

Part 3: Write the inverse, then verify by composition.

11. $f(x) = (x-2)^3 + 1$

Write the inverse: $f^{-1}(x) =$ ______________________________

Verify by showing $f(f^{-1}(x)) = x$:

12. $f(x) = 2\sqrt[3]{x+3} - 4$

Write the inverse: $f^{-1}(x) =$ ______________________________

Verify by showing $f(f^{-1}(x)) = x$:

Part 4: Quick Write

13. In your own words, explain why $f(f^{-1}(x)) = x$ makes sense. What does it mean for a function and its inverse to "undo" each other?